Advanced Topics in Learning and Games - Project Proposal

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1 Problem Introduction

In [4] it was proven that for general-sum multiplayer finite games, if all players are running the same Optimistic Hedge algorithm, they will suffer a regret of $\mathcal{O}(\log^4(T))$.

In [6] the result was generalized to convex games and a wider family of algorithms, with the better bound of $\mathcal{O}(\log(T))$. As a special case, there are the Optimistic Hedge algorithm mentioned before and the Extra Gradient method. As a consequence, it means we can approximate a CCE at a much faster rate than the classic $\mathcal{O}(\sqrt{T})$ bound of the adversarial settings.

However, it remains an open question whether these upper bounds are actually tight. Does there exist a pathological general-sum game geometry that mathematically forces this logarithmic growth (e.g., by trapping the algorithms in chaotic limit cycles), or is the true worst-case regret bounded by a tighter function, or even a constant $\mathcal{O}(1)$? Furthermore, the hidden constants in the Big-$\mathcal{O}$ notation may scale poorly with the number of actions, causing massive transient regret before the asymptotic rate kicks in.

In this project, we will answer these questions by simulating these algorithms and analyzing their regret behavior over large horizons of T. Rather than relying on sampling random games, we will utilize optimization techniques to actively discover the payoff matrices that maximize the algorithms' regret for large values of T, empirically testing the theoretical lower bounds.

As an extension, we will adapt the DGDA method [8] to the simplex and also analyze its asymptotic behaviour.

2 Approach

2.1 Worst Game Optimization

Theoretically, because the game dynamics are known, deterministic, and differentiable, we may use a standard differentiation library like autograd to calculate the gradient with respect to the utility matrices and optimize. In practice, for large horizons of T it is not practical, both because it will take a large amount of RAM, and because of the exploding/vanishing gradient problem.

Instead, we may use 0th-order optimization methods, such as the CMA-ES [7] method, which is considered state-of-the-art. While this method may have good results for a smaller number of actions, this algorithm (and 0th-order optimization methods in general) suffers from the curse of dimensionality, due to the exponentially growing search space.

Therefore, in addition to using the CMA-ES method, we will apply optimization over a smooth ODE that approximates the discrete learning dynamics. In doing so, we rely on the works of [1, 3, 5] that showed how to formulate such learning dynamics as ODEs, and on the work of [2] which shows how to optimize such ODEs with respect to parameters such as our utility matrices, using ODE solvers in terms of time and memory, bypassing the risk of vanishing or exploding gradients.

2.2 Asymptotic Behaviour Analysis

We will apply several analysis methods, such as visualization with coordinate scaling (checking whether the scaled graph forms a line), curve fitting using linear regression, and derivative analysis-based methods.

In addition, in order to analyze and visualize the strategy behaviour, we will plot the path of the mean and the last iterate strategies derived from different algorithms, and analyze their behaviour (does it converge? does it get stuck in a cycle?)

3 Timeline

1. Establish a framework for running learning dynamics such as OMWU and EG for long horizons, for a specific game
2. Implement CMA-ES optimization loop and verify it against random games
3. Establish a framework for analyzing the asymptotic behaviour and strategy convergence for a specific game (plots, linear regression, derivative analysis)
4. Formulate the continuous-time Modified Differential Equations for the algorithms. Integrate a differential equation solver and validate the method against the CMA-ES method and random games.
5. Scale the experiments to games with a larger number of actions
6. Formulate the DGDA algorithm for the simplex and as an ODE, and analyze its behaviour using the established framework
7. Compile the empirical results, visualizations, and theoretical insights into the final project manuscript

References

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